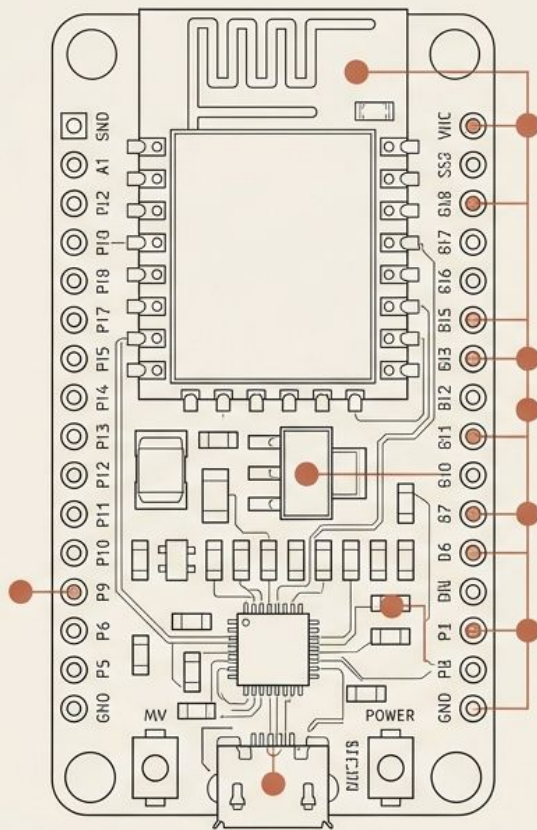




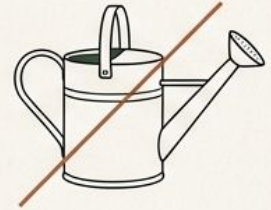
Giving Nature a Voice and an Internet Connection

The Autonomous Smart Garden:
An IoT-driven agricultural
ecosystem powered by the
ESP8266.

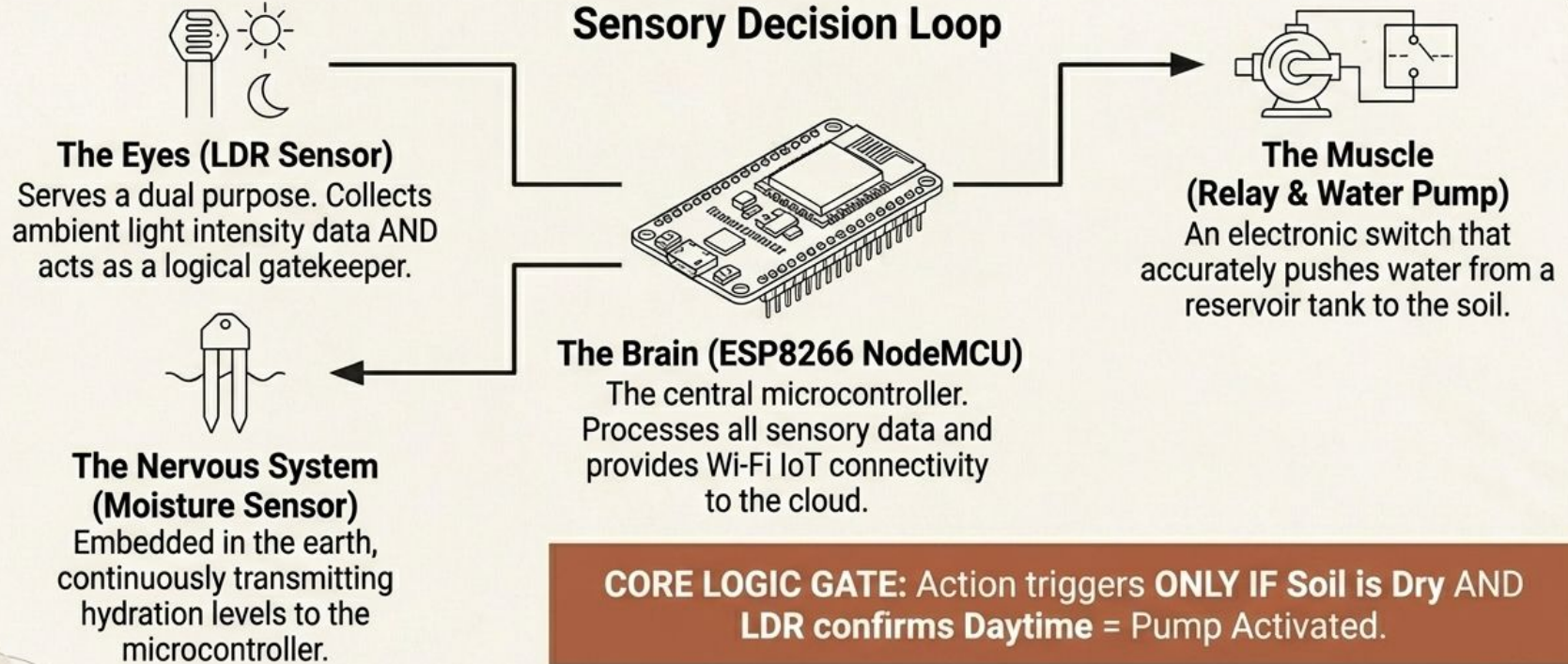
*Plants cannot speak. We gave
them a sensor network and Wi-Fi.*

Shifting from Reactive Guesswork to Proactive Automation

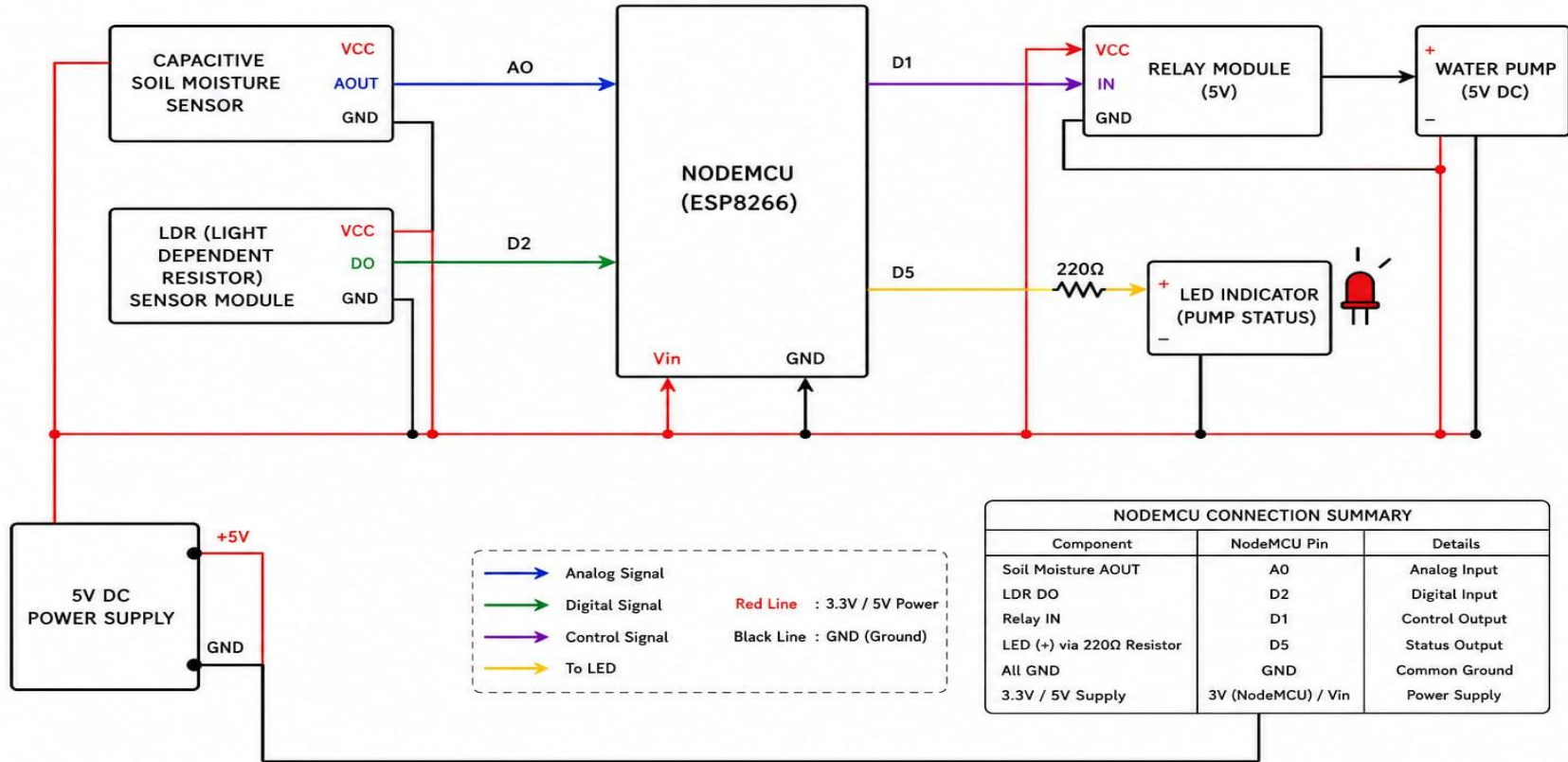
	Traditional Plant Care	The Autonomous Smart Garden
Watering Logic	Manual guesswork; high risk of overwatering or forgetting entirely.	Automated precision triggered strictly by continuous, real-time soil moisture monitoring.
Environmental Awareness	Completely blind to day/night cycles and light exposure.	Light-aware logic prevents watering at night, averting root rot and mimicking natural ecosystems.
Remote Management	Plants left unmonitored during college trips or holidays.	Complete IoT control and telemetry monitoring from a smartphone, accessible globally.



The Sensory Anatomy of an Autonomous Plant



IoT-BASED SMART PLANT WATERING SYSTEM



Complete Visibility and Control from Anywhere

Live Telemetry

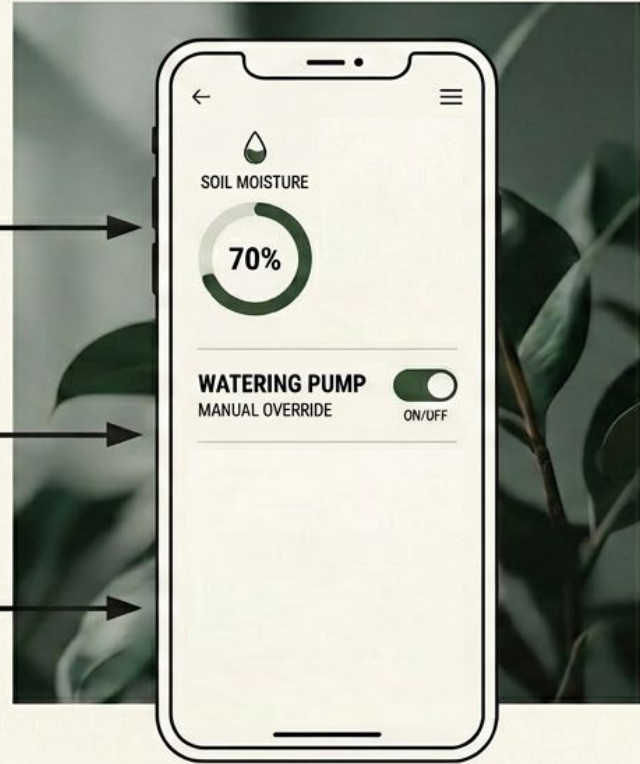
Real-time data streams displaying current Soil Moisture levels and Live Light Intensity gathered directly from the LDR.

Remote Actuation

A manual pump override toggle switch, allowing the user to bypass automation.

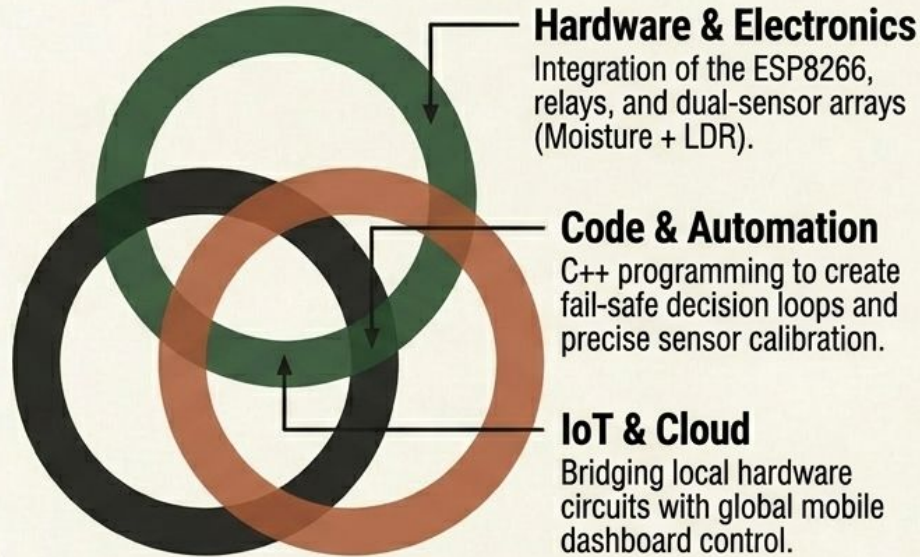
Connection Status

Shows the real-time connectivity to the cloud, indicating if the data is up-to-date.



Imagine being away at college all day. Your plant is continuously monitored, watered only when necessary, and fully visible in your pocket.

Beyond Blinking LEDs: A Foundation for Smart Agriculture



Future Scalability Roadmap

Phase Two Architecture

- ☒ Water level monitoring in the reservoir tank.
- ☒ Historical moisture and light data logging.
- ☒ Weather API integration for rain delays.
- ☒ AI-driven predictive watering models.
- ☒ Multi-plant network scaling.

“More than a student build—a genuine smart agriculture system.”